

Design and Closed-Loop Analysis of a 389 V DC Input, 24 V / 15 A Output Isolated Two-Switch Forward Converter

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1. Objective of the Project

The objective of this study is to develop a power conversion structure suitable for loads that require low voltage and high current, supplied from a high-voltage DC source. Within this scope, a galvanically isolated Two-Switch Forward converter rated at 360 W, converting 389 V DC input to 24 V / 15 A output, has been designed and analyzed under closed-loop control.

The primary motivation of the project is to present an end-to-end design procedure for a reliable, isolated, and well-regulated DC–DC power stage intended for high input voltage applications. Accordingly, the study addresses not only the fundamental voltage conversion relationships but also magnetic component design, semiconductor voltage and current stresses, conduction and switching losses, output filter sizing, and compensation network design as integral parts of the overall development process.

At the conclusion of the study, the suitability of the selected topology for the 360 W power level has been evaluated. The key converter waveforms, output voltage regulation performance, and major component sizing have been verified through simulation results. Thus, the project aims to provide a comprehensive presentation of both the theoretical framework and practical engineering steps involved in the design of high-power isolated DC–DC converters.

2. Rationale for Selecting the Two-Switch Forward Topology

In this study, the objective is to obtain an isolated 24 V / 15 A output at a total power level of 360 W from a 389 V DC input source. In line with these requirements, various DC–DC converter topologies were evaluated, and the Two-Switch Forward configuration was selected based on the required power level, isolation requirement, and semiconductor voltage stress considerations.

The flyback topology is widely used in low- and medium-power applications due to its simplicity and minimal component count. However, as the power level approaches 360 W, transformer current stress, switch voltage stress, and output ripple increase significantly. For this reason, the flyback topology was deemed unsuitable for the present design.

Although the buck converter offers structural simplicity and high efficiency, it does not provide galvanic isolation. Since isolation is a fundamental requirement of this project, the buck topology was excluded from consideration.

In contrast, the Two-Switch Forward topology provides a more suitable solution for medium- to high-power applications. The principal advantages of this configuration are summarized as follows:

- More appropriate energy transfer characteristics at the 360 W power level compared to the flyback topology
- Reduced and better-controlled voltage stress across the primary switches
- Transformer core reset capability without requiring an additional reset winding
- Inherent fulfillment of the galvanic isolation requirement
- Lower current and voltage ripple due to the presence of an output filter

For these reasons, the Two-Switch Forward topology was determined to be the most technically appropriate choice for this high-input-voltage, isolated-output application.

3. Design Specifications and Key Technical Parameters

The fundamental electrical requirements and design targets used in the converter design are summarized in Table 1. These parameters constitute the basis for power stage dimensioning, magnetic component design, semiconductor selection, and control structure development.

Table 1. Design specifications and technical parameters

Parameter	Symbol	Value
Minimum DC bus voltage	$V_{DC(min)}$	224.3 V
Maximum DC bus voltage	$V_{DC(max)}$	389 V
Output voltage	V_o	24 V
Output current	I_o	15 A
Output power	P_o	360 W
Target efficiency	η	90%
Input power	P_{in}	400 W
Switching frequency	f_s	100 kHz
Maximum duty cycle	D_{max}	0.45
Minimum duty cycle	D_{min}	0.237
Output current ripple target	ΔI_o	3 A
Topology	—	Two-Switch Forward
Isolation	—	Yes
Control structure	—	Closed-loop

Based on these technical specifications, the fundamental conversion ratio, transformer turns ratio, output filter design, and semiconductor device dimensioning were carried out accordingly.

4. Fundamental Converter Design

The first step of the design process is to determine the DC bus voltage limits formed after rectification. Although many reference schematics represent the input as an ideal voltage source, in a practical implementation a DC-link (bulk) capacitor is mandatory to suppress the ripple of the rectified DC voltage. This capacitor supplies energy to the circuit when the mains voltage approaches zero and therefore determines the minimum DC bus voltage.

Assuming an output power of $P_o = 360$ W and a system efficiency of $\eta = 0.90$, the input power and total power loss are calculated as

$$P_{in} = \frac{P_o}{\eta} = \frac{360}{0.90} = 400 \text{ W}$$

$$P_{loss} = P_{in} - P_o = 400 - 360 = 40 \text{ W}$$

The minimum DC bus voltage $V_{DC(min)}$ is calculated by considering the input capacitance and the line ripple effect. For this calculation, a standard bulk capacitor value of 470 μ F is assumed, consistent with the design power level

$$V_{DC(min)} = \sqrt{2 \cdot V_{in,rms}^2 - \frac{P_{in} \cdot t_{discharge}}{C_{in} \cdot f_{line}}}$$

Substituting the parameters:

$$V_{DC(min)} = \sqrt{2 \cdot (180)^2 - \frac{400 \cdot (1 - 0.2)}{470 \times 10^{-6} \cdot 100}}$$

$$= 224.3 \text{ V}$$

The maximum DC bus voltage $V_{DC(max)}$ corresponds to the peak value of the maximum line voltage:

$$V_{DC(max)} = \sqrt{2} \cdot V_{in,max} = \sqrt{2} \cdot 275 = 389 \text{ V}$$

The transformer turns ratio n is determined based on the selected primary and secondary turns, $N_p = 30$ and $N_s = 8$:

$$n = \frac{N_p}{N_s} = \frac{30}{8} = 3.75$$

The duty cycle limits are calculated according to the selected voltage range and turns ratio.

Minimum Duty Cycle:

$$D_{min} = \frac{V_o + V_f}{V_{DC(max)}} \cdot n = \frac{24 + 0.65}{389} \cdot 3.75 = 0.237$$

Maximum Duty Cycle (Design Limit):

$$D_{max} = 0.45$$

Based on these values, the converter has been dimensioned to operate safely over an input voltage range of 224.3 V to 389 V, with a duty cycle variation between 0.237 and 0.45.

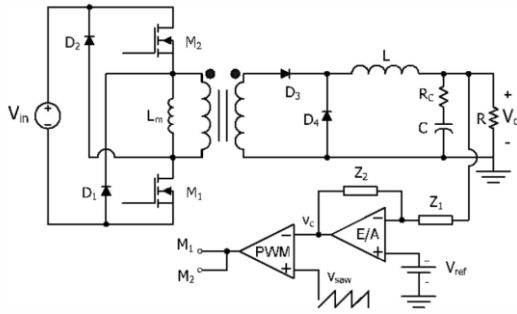


Figure 1. Type III Compensator with Gain Curve

5. Transformer and Magnetic Design

A high-frequency transformer is employed in the converter to ensure galvanic isolation and energy transfer. An ETD44 core has been selected for this design. The primary motivation for this choice is its ability to provide sufficient magnetic capacity at a power level of 360 W and a switching frequency of 100 kHz. The ETD44 geometry is widely used in switched-mode power supplies within this power range, offering an appropriate balance between power density and thermal performance.

Another important reason for selecting the ETD44 core is its effective cross-sectional area and window area. According to the datasheet, the effective core cross-sectional area is

$$A_e = 173 \text{ mm}^2$$

This value allows the primary number of turns to remain within a reasonable range. If a smaller core were used, a higher number of primary turns would be required to operate within the same flux density

limits, leading to increased copper losses and greater winding complexity.

Since the converter provides a 24 V / 15 A output on the secondary side, the conductor cross-section becomes critical due to the high output current. Considering the skin effect at 100 kHz, the secondary winding must be implemented using either litz wire or multiple parallel conductors. Therefore, a core structure with sufficient window area to physically accommodate the windings is required. In this regard, the ETD44 provides a more suitable solution compared to smaller core alternatives.

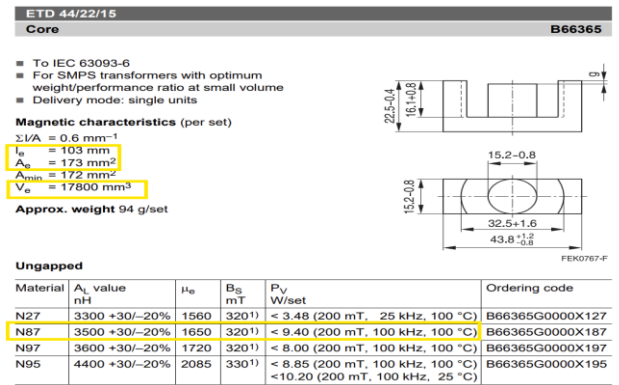


Figure 2. ETD 44/22/15 Datasheet Specifications

The core material selected is N87 ferrite. N87 is suitable for switched-mode power supplies operating at 100 kHz due to its favorable core loss characteristics. Additionally, the datasheet provides

$$A_L = 3500 \text{ nH/turn}^2$$

which enables direct calculation of the magnetizing inductance.

The allowable flux density variation in the design is specified as

$$\Delta B = 0.2 \text{ T}$$

The minimum required primary turns are calculated using

$$N_{p(min)} = \frac{V_{DC(min)} \cdot D_{max}}{A_e \cdot f_s \cdot \Delta B} \times 10^6$$

Substituting the numerical values:

$$N_{p(min)} = \frac{224.3 \cdot 0.45 \cdot 10^6}{173 \cdot 100000 \cdot 0.2} = 29.17$$

Accordingly, the primary number of turns is selected as

$$N_p = 30$$

The actual flux density variation with the selected primary turns is calculated as

$$\Delta B = \frac{V_{DC(min)} \cdot D_{max} \cdot 10^6}{A_e \cdot f_s \cdot N_p}$$

$$\Delta B = \frac{224.3 \cdot 0.45 \cdot 10^6}{173 \cdot 100000 \cdot 30} = 0.134 \text{ T}$$

$$\Delta B = 134 \text{ mT}$$

Since this value is below the allowable limit of 200 mT, core saturation is avoided under worst-case operating conditions.

The secondary turns are determined based on the specified turns ratio:

$$N_s = \frac{N_p}{n} = \frac{30}{3.75} = 8$$

The magnetizing inductance is calculated using

$$L_m = A_L \cdot N_p^2$$

$$L_m = 3500 \cdot 30^2 \text{ nH} = 3.15 \text{ mH}$$

The corresponding maximum magnetizing current is given by

$$I_{mag(max)} = \frac{V_{DC(min)} \cdot D_{max}}{L_m \cdot f_s}$$

$$I_{mag(max)} = \frac{224.3 \cdot 0.45}{3.15 \times 10^{-3} \cdot 100000} = 0.32 \text{ A}$$

In conclusion, the ETD44/N87 core provides an appropriate magnetic solution for this design in terms of power level, operating frequency, winding accommodation, and flux density limits.

Table 2. Transformer specifications.

Magnetic Parameter	Value
Core type	ETD44
Material	N87

A_e	173 mm ²
A_L	3500 nH/turn ²
N_p	30
N_s	8
L_m	3.15 mH
ΔB	134 mT

6. Output Stage Design

In the output stage design, the output inductance and capacitance were dimensioned to maintain continuous conduction mode (CCM) operation and to limit the output voltage ripple. The output current ripple was selected as 20% of the nominal load current. Accordingly,

$$\Delta I_o = 0.2 \cdot I_o = 0.2 \cdot 15 = 3 \text{ A}$$

The minimum output inductance was calculated under maximum input voltage and minimum duty cycle conditions using the following expression:

$$L_{min} = \left(\frac{V_{DC(max)}}{n} - (V_o + V_f) \right) \cdot \frac{D_{min}}{f_s \cdot \Delta I_o}$$

Substituting the numerical values:

$$L_{min} = \left(\frac{389}{3.75} - (24 + 0.65) \right) \cdot \frac{0.237}{100000 \cdot 3}$$

$$L_{min} \approx 68 \mu\text{H}$$

Accordingly, the output inductor was selected as

$$L_o = 68 \mu\text{H}$$

The output capacitor value was chosen as

$$C_o = 1000 \mu\text{F}$$

with an equivalent series resistance

$$ESR = 20 \text{ m}\Omega$$

The output voltage ripple, considering both capacitive and ESR effects, was calculated using:

$$\Delta V_o = \Delta I_o \left(ESR + \frac{1}{4C_o f_s} \right)$$

Substituting the numerical values:

$$\Delta V_o = 3 \left(0.02 + \frac{1}{4 \cdot 1000 \times 10^{-6} \cdot 100000} \right)$$

$$\Delta V_o = 67.5 \text{ mV}$$

In simulation, the output capacitor current was observed to vary approximately within the range of ± 0.3125 A. This indicates that the output filter effectively suppresses current ripple and that the capacitor current stress remains within acceptable limits.

In conclusion, with the selected output filter parameters, continuous conduction mode is maintained, low output voltage ripple is achieved, and a suitable filter structure is obtained for the 24 V / 15 A output specification.

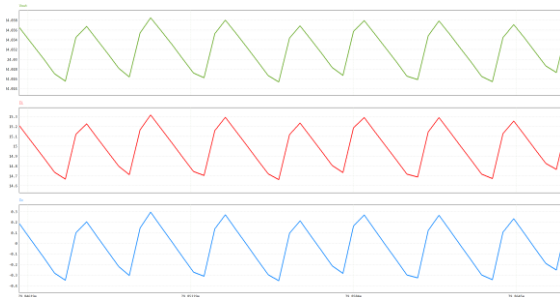


Figure 3. Output Voltage, Inductor Current, and Capacitor Current

7. Semiconductor Stress and Loss Analysis

In this section, the primary voltage and current stresses of the two MOSFETs on the primary side and the rectifier diodes on the secondary side of the two-switch forward converter are analyzed. Subsequently, the dominant conduction and switching losses are calculated.

7.1. Primary-Side MOSFET Voltage Stress

In the two-switch forward topology, two MOSFETs operate simultaneously on the primary side. One of the main advantages of this configuration is that the maximum voltage stress across each MOSFET is

approximately limited to the DC bus voltage. Therefore, the primary selection criterion for the MOSFET is $V_{DS(max)} \geq V_{DC(max)}$

Since the maximum DC bus voltage is

$$V_{DC(max)} = 389 \text{ V}$$

the MOSFET breakdown voltage must be selected with an adequate safety margin above this value. Considering a safety factor k_M , the following relation may be used:

$$V_{DC(max)} = k_M \cdot BVDSS$$

Accordingly, MOSFETs with a voltage rating of 500 V or higher are considered appropriate for this application.

7.2. Primary-Side Current Stresses

The primary current stress reaches its maximum under minimum input voltage conditions. The average primary current is calculated as

$$I_{p(avg)} = \frac{P_{in}}{V_{DC(min)}}$$

$$I_{p(avg)} = \frac{400}{224.3} = 1.78 \text{ A}$$

The average primary current during the switch conduction interval is

$$I_{p(on)} = \frac{P_{in}}{V_{DC(min)} \cdot D_{max}}$$

$$I_{p(on)} = \frac{400}{224.3 \cdot 0.45} = 3.96 \text{ A}$$

The maximum magnetizing current due to the transformer magnetizing inductance is

$$I_{mag(max)} = \frac{V_{DC(min)} \cdot D_{max}}{L_m \cdot f_s}$$

$$I_{mag(max)} = \frac{224.3 \cdot 0.45}{3.15 \times 10^{-3} \cdot 100000} = 0.32 \text{ A}$$

The primary peak current is expressed as the sum of the reflected load current component and the magnetizing current:

$$I_{p(peak)} = I_{p(on)} + \frac{\Delta I_o}{2n} + I_{mag(max)}$$

$$I_{p(peak)} = 3.96 + \frac{3}{2 \cdot 3.75} + 0.32 = 4.68 \text{ A}$$

The calculated primary RMS current is

$$I_{p(rms)} = 2.87 \text{ A}$$

This value is used for determining the MOSFET conduction losses.

7.3 MOSFET Loss Analysis

The total power loss of a single MOSFET consists of conduction loss, turn-on loss, and turn-off loss:

$$P_{MOS} = P_{cond} + P_{sw(on)} + P_{sw(off)}$$

Conduction Loss

$$P_{cond} = I_{p(rms)}^2 \cdot R_{DS(on)}$$

$$R_{DS(on)} = 0.28 \Omega$$

$$P_{cond} = (2.87)^2 \cdot 0.28 = 2.30 \text{ W}$$

Turn-On Loss

$$P_{sw(on)} = \frac{1}{2} I_{p(valley)} \cdot V_{bulk} \cdot \Delta t \cdot f_s$$

For the present design:

$$P_{sw(on)} = 0.49 \text{ W}$$

Turn-Off Loss

$$P_{sw(off)} = \frac{1}{6} I_{p(peak)} \cdot V_{bulk,max} \cdot \Delta t \cdot f_s$$

$$P_{sw(off)} = 1.21 \text{ W}$$

Thus, the total loss for a single MOSFET is

$$P_{MOS} = 2.30 + 0.49 + 1.21 = 4.0 \text{ W}$$

Since two MOSFETs are used on the primary side:

$$P_{MOS(total)} = 2 \cdot P_{MOS} = 2 \cdot 4.0 = 8.0 \text{ W}$$

7.4. Secondary Diode Voltage Stress

The secondary side includes a forward rectifier diode and a freewheeling diode. The approximate maximum reverse voltage stress is calculated using the turns ratio and maximum input voltage:

$$V_R = \frac{V_{DC(max)}}{n} \cdot 1.5$$

$$V_R = \frac{389}{3.75} \cdot 1.5 = 155.6 \text{ V}$$

Accordingly, diodes with a voltage rating of 200 V are considered suitable.

7.5. Secondary Diode Current Stresses

The average current of the forward rectifier diode is:

$$I_{D1(avg)} = I_o \cdot D_{max}$$

$$I_{D1(avg)} = 15 \cdot 0.45 = 6.75 \text{ A}$$

The average current of the freewheeling diode is:

$$I_{D2(avg)} = I_o \cdot (1 - D_{min})$$

$$I_{D2(avg)} = 15 \cdot (1 - 0.237) = 11.45 \text{ A}$$

These values are used in determining the appropriate current ratings of the diodes.

7.6. Diode Losses

Assuming a forward voltage drop of

$$V_f = 0.65 \text{ V}$$

Forward Rectifier Diode Loss

$$P_{D1} = V_f \cdot I_o \cdot D_{max}$$

$$P_{D1} = 0.65 \cdot 15 \cdot 0.45 = 4.38 \text{ W}$$

Freewheeling Diode Loss

$$P_{D2} = V_f \cdot I_o \cdot (1 - D_{min})$$

$$P_{D2} = 0.65 \cdot 15 \cdot (1 - 0.237) = 7.43 \text{ W}$$

7.7 Transformer Loss Analysis

Transformer losses directly affect overall efficiency in isolated converter designs. In this section, copper losses and core losses are considered separately. All calculations are performed under worst-case operating conditions:

$$V_{DC(min)} = 224.3 \text{ V}, D_{max} = 0.45$$

7.7.1 Conductor Cross-Section Selection

In high-frequency transformer windings, conductor cross-section is determined according to the current density criterion. A typical design value for switched-mode power supplies is

$$J = 4 \text{ A/mm}^2$$

For the secondary winding:

$$A_{wire,p} = \frac{I_{p(rms)}}{J} = \frac{2.87}{4} = 0.718 \text{ mm}^2$$

The closest standard wire size, AWG 18 ($A_{wire} = 0.823 \text{ mm}^2$, $d = 1.024 \text{ mm}$), has been selected.

For the secondary winding, considering the 15 A output current and the high RMS current value ($I_{s,rms} = 8.208 \text{ A}$), the required conductor cross-section is calculated as

$$A_{wire,s} = \frac{I_{s(rms)}}{J} = \frac{8.208}{4} = 2.052 \text{ mm}^2$$

A standard 2.5 mm² multi-stranded (Litz) conductor has been selected. In order to reduce the skin effect at the operating frequency of 100 kHz, the use of Litz wire is adopted for the secondary winding.

7.7.2 Winding DC Resistance Calculation

The DC resistance of the windings is calculated using

$$R_{dc} = \rho \cdot \frac{N \cdot l_N}{A_{wire}}$$

where ρ is the resistivity of copper, N is the number of turns, l_N is the mean length per turn, and A_{wire} is the conductor cross-sectional area.

For the ETD44 bobbin, the mean turn length is taken from the manufacturer's data as

$$l_N = 77.7 \text{ mm}$$

The resistivity of copper at an operating temperature of 100°C is assumed as

$$\rho = 2.1 \times 10^{-8} \Omega \cdot \text{m}$$

For the primary winding ($N_p = 30$, $A_{wire,p} = 0.823 \times 10^{-6} \text{ m}^2$):

$$R_{dc,p} = 2.1 \times 10^{-8} \cdot \frac{30 \times 0.0777}{0.823 \times 10^{-6}}$$

$$= 2.1 \times 10^{-8} \cdot \frac{2.331}{0.823 \times 10^{-6}}$$

$$R_{dc,p} = 0.0595 \Omega$$

For the secondary winding ($N_s = 8$, $A_{wire,s} = 2.5 \times 10^{-6} \text{ m}^2$):

$$R_{dc,s} = 2.1 \times 10^{-8} \cdot \frac{8 \times 0.0777}{2.5 \times 10^{-6}}$$

$$= 2.1 \times 10^{-8} \cdot \frac{0.6216}{2.5 \times 10^{-6}}$$

$$R_{dc,s} = 0.00522 \Omega$$

7.7.3 Skin Effect and AC Resistance Evaluation

At high frequencies, current tends to flow near the surface of the conductor, which reduces the effective cross-sectional area and increases resistance. For copper at 100°C, the penetration depth is calculated using the empirical expression [4]:

$$\delta = \frac{7.6}{\sqrt{f}} \text{ cm}$$

At a switching frequency of 100 kHz:

$$\delta = \frac{7.6}{\sqrt{100000}} = \frac{7.6}{316.2} = 0.024 \text{ cm}$$

$$= 0.24 \text{ mm}$$

The diameter of the AWG 18 conductor used in the primary winding ($d = 1.024 \text{ mm}$) is approximately 4.3 times the penetration depth. This indicates that the skin effect is significant. However, considering that the primary winding is implemented as a single layer within the ETD44 window and that Litz wire may be used, the ratio

$$R_{ac}/R_{dc} \approx 1.5$$

is assumed.

For the secondary winding, due to the use of Litz wire at 100 kHz, skin and proximity effects are substantially reduced. Therefore,

$$R_{ac}/R_{dc} \approx 1.1$$

is assumed for the secondary.

These coefficients correspond to typical values derived from Dowell's method for single-layer windings and Litz wire assumptions.

7.7.4 Total Copper Loss

The total copper loss, including both DC and AC components, is expressed as

$$P_{cu} = I_{rms}^2 \cdot R_{dc} \cdot \frac{R_{ac}}{R_{dc}}$$

Primary copper loss:

$$\begin{aligned} P_{cu,p} &= (2.87)^2 \times 0.0595 \times 1.1 \\ &= 8.237 \times 0.0595 \times 1.1 \\ P_{cu,p} &= 0.735 \text{ W} \end{aligned}$$

Secondary copper loss:

$$\begin{aligned} P_{cu,s} &= (8.208)^2 \times 0.00522 \times 1.1 \\ &= 67.37 \times 0.00522 \times 1.1 \\ P_{cu,s} &= 0.387 \text{ W} \end{aligned}$$

Total copper loss:

$$P_{cu(total)} = 0.735 + 0.387 = 1.122 \text{ W}$$

7.7.5 Core Loss

Core loss consists of hysteresis and eddy current losses. According to the manufacturer's data for N87 ferrite material, under 200 mT and 100 kHz conditions, the loss per ETD44 core set is specified as

$$P_V < 9.40 \text{ W}$$

In this study, the calculated flux density variation is

$$\Delta B = 134 \text{ mT}$$

which is below the 200 mT reference value provided by the manufacturer. Since the forward converter operates with unipolar magnetization, the exact core loss depends on the operating conditions and the B-H characteristics of the material.

Based on the manufacturer's core loss data and power density curves, the core loss under the present operating conditions is estimated as

$$P_{core} \approx 2.0 \text{ W}$$

This value is obtained by scaling the 200 mT reference loss to 134 mT and considering the effect of unipolar magnetization. Experimental validation is recommended for precise determination.

7.7.6 Total Transformer Loss

Table 3. Transformer Loss Summary

Loss Component	Value (W)
Primary copper loss $P_{cu,p}$	0.735
Secondary copper loss $P_{cu,s}$	0.387
Core loss P_{core}	2.0
Total transformer loss	3.122

7.8 Total Loss Distribution and Efficiency Estimation

All converter loss components are combined to determine the total loss and system efficiency.

Table 4. Total Loss Distribution

Loss Source	Loss (W)	Ratio (%)
Primary MOSFET conduction (2×)	4.60	19.8

Loss Source	Loss (W)	Ratio (%)
Primary MOSFET switching (2×)	3.40	14.6
Secondary diode — forward (D ₁)	4.38	18.8
Secondary diode — freewheeling (D ₂)	7.43	31.9
Transformer copper loss	1.12	4.8
Transformer core loss	2.0	8.6
Other (gate drive, parasitics, etc.)	~0.35	1.5
Total loss	23.28	100

The system efficiency is calculated as

$$\eta = \frac{P_o}{P_o + P_{loss}} = \frac{360}{360 + 23.28}$$

$$\eta = \frac{360}{383.28} \approx 93.9\%$$

7.9. Evaluation

At the beginning of the design, the target efficiency was specified as $\eta = 90\%$, and the input power was assumed as $P_{in} = 400$ W accordingly. Based on the detailed loss analysis, the calculated efficiency of approximately 93.9% exceeds the initial target and indicates that the design operates with sufficient margin in terms of the loss budget.

The loss distribution reveals that the largest contribution originates from the secondary rectification diodes. The freewheeling diode (D₂) alone accounts for approximately 32% of the total loss, indicating that this component is the primary candidate for efficiency improvement.

In future work, the implementation of synchronous rectification is planned in order to reduce diode conduction losses to the level of MOSFET channel resistance, with the objective of increasing the overall efficiency to 96% or higher.

8. Closed-Loop Control Structure

In order to maintain a constant output voltage despite variations in input voltage and load conditions, a closed-loop control structure is employed. In open-loop operation, the duty cycle remains fixed; therefore, input and load variations directly affect the output voltage. For this reason, a feedback-based voltage control scheme is adopted.

The output voltage is sensed through a resistive voltage divider and compared with a reference voltage. The feedback relationship is given by

$$V_{fb} = V_o \cdot \frac{R_4}{R_1 + R_4}$$

For $V_{ref} = 2.5$ V and $R_4 = 10$ k Ω , the upper resistor is selected as

$$R_1 = 10k \cdot \left(\frac{24}{2.5} - 1 \right) = 86 \text{ k}\Omega$$

Thus, at the nominal output voltage, the feedback voltage is adjusted to approximately 2.5 V.

The control structure is implemented based on voltage-mode PWM. The output of the error amplifier is compared with a 100 kHz sawtooth carrier signal to generate the switching signals. In this way, the output voltage error is converted into a duty cycle variation, providing closed-loop regulation.

Since the output LC filter operating in continuous conduction mode exhibits a double-pole characteristic, a Type-III compensation network is employed to compensate for phase lag and to ensure adequate phase margin. The resonance frequency of the output filter is calculated as

$$f_{LC} = \frac{1}{2\pi\sqrt{L_o C_o}}$$

For $L_o = 68$ μ H and $C_o = 1000$ μ F:

$$f_{LC} = \frac{1}{2\pi\sqrt{68 \times 10^{-6} \cdot 1000 \times 10^{-6}}} \approx 611 \text{ Hz}$$

Using the op-amp-based Type-III compensator model provided in the TI application note, and assuming $C_1 \gg C_3$, the pole and zero frequencies are defined as follows:

$$\begin{aligned} f_{p0} &= \frac{1}{2\pi R_1(C_1 + C_3)} \approx \frac{1}{2\pi R_1 C_1} \\ f_{p1} &= \frac{1}{2\pi R_3 C_2} \\ f_{p2} &= \frac{1}{2\pi R_2 \left(\frac{C_1 C_3}{C_1 + C_3} \right)} \approx \frac{1}{2\pi R_2 C_3} \\ f_{z1} &= \frac{1}{2\pi(R_1 + R_3)C_2} \\ f_{z2} &= \frac{1}{2\pi R_2 C_1} \end{aligned}$$

Accordingly, the passive components are determined from

$$\begin{aligned} C_1 &= \frac{f_{p2} - f_{z2}}{2\pi R_1 f_{p0} f_{p2}} \\ C_2 &= \frac{f_{p1} - f_{z1}}{2\pi R_1 f_{p1} f_{z1}} \\ C_3 &= \frac{f_{z2}}{2\pi R_1 f_{p0} f_{p2}} \\ R_2 &= \frac{R_1 f_{p0} f_{p2}}{(f_{p2} - f_{z2}) f_{z2}} \\ R_3 &= \frac{R_1 f_{z1}}{f_{p1} - f_{z1}} \end{aligned}$$

In this study, the compensation network component values are selected approximately as

$$\begin{aligned} R_1 &= 86 \text{ k}\Omega, R_4 = 10 \text{ k}\Omega \\ R_2 &= 9.6 \text{ k}\Omega, R_3 = 373 \Omega \\ C_1 &= 77 \text{ nF}, C_2 = 8.5 \text{ nF}, C_3 = 331 \text{ pF} \end{aligned}$$

The crossover frequency is targeted at approximately 10 kHz. In this way, the control loop bandwidth is maintained sufficiently below the 100 kHz switching frequency, ensuring stable operation and fast dynamic response. Simulation results show that the feedback voltage stabilizes at approximately 2.503 V, confirming that the closed-loop control structure provides accurate regulation at the nominal operating point.

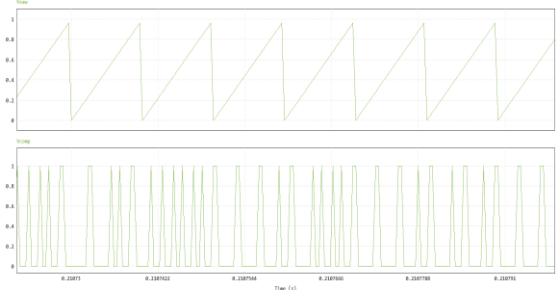


Figure 4. PWM Waveforms

In the PWM modulation structure, a 100 kHz sawtooth carrier signal is used. The comparator output obtained by comparing the carrier signal with the control voltage determines the duty cycle of the converter, thereby providing closed-loop voltage regulation.

9. Simulation Results and Verification

The designed two-switch forward converter has been verified through simulation under closed-loop control at nominal operating conditions. In this section, the output voltage and current, primary and secondary currents, output filter currents, and the fundamental waveforms related to PWM generation are evaluated.

Under nominal operating conditions, the input voltage is applied as 389 V. Simulation results show that the output voltage is regulated at approximately 24 V. The average output voltage is obtained as

$$V_{out(avg)} \approx 24.03 \text{ V}$$

and the average output current is obtained as

$$I_{out(avg)} \approx 15.02 \text{ A}$$

These results indicate that the targeted 24 V / 15 A operating point has been successfully achieved.

A low-amplitude ripple is observed in the output voltage waveform. This ripple level confirms that the selected output inductor and capacitor values are adequate. The load current remains approximately constant under nominal conditions, demonstrating that the closed-loop regulation operates correctly.

When the primary and secondary winding currents are examined, the waveforms are observed to be pulsed in a manner consistent with the forward

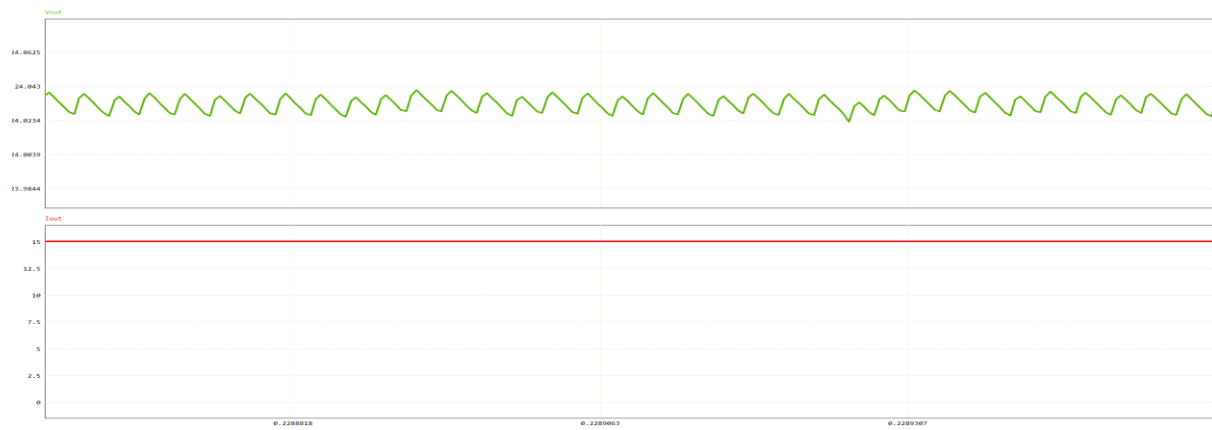


Figure 6. Output voltage and output current waveforms under nominal load.

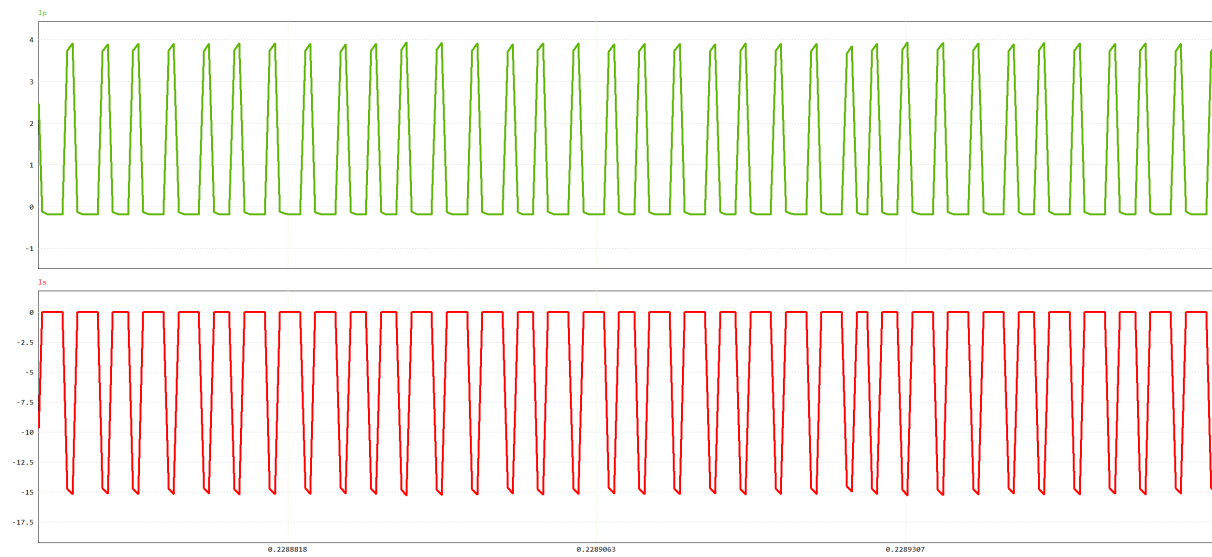


Figure 7. Primary and secondary winding currents.

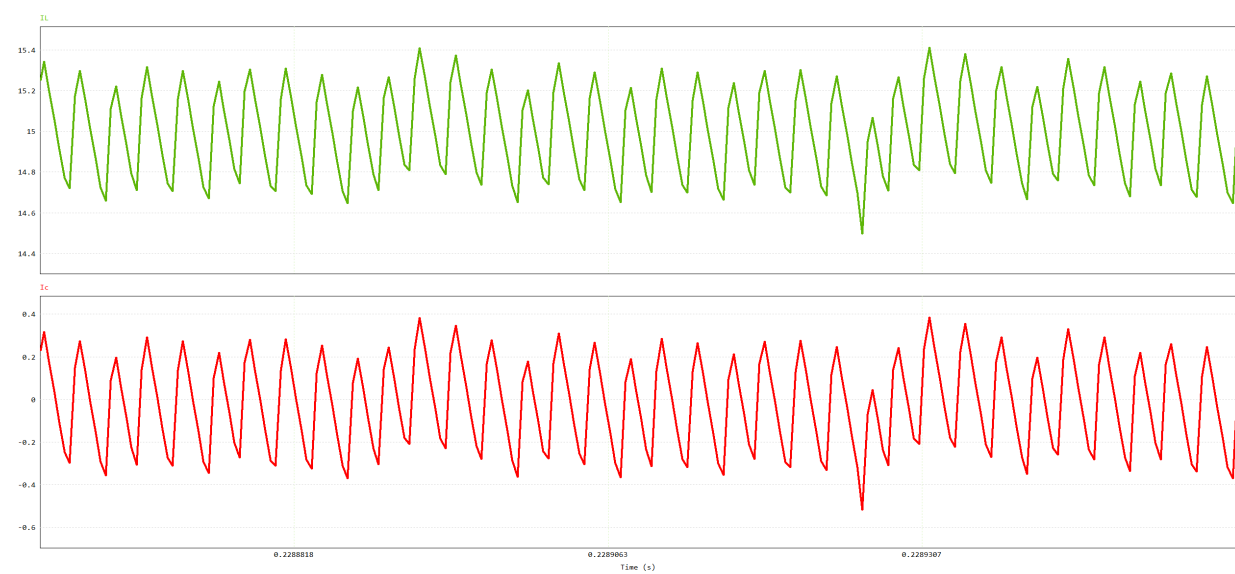


Figure 8. Output inductor and capacitor currents.

10. Conclusion and Evaluation

In this study, a 360 W isolated two-switch forward converter providing a 24 V / 15 A output from a 389 V DC input has been designed and analyzed under closed-loop operation. Based on the theoretical calculations, the transformer, output filter, semiconductor devices, and compensation network have been dimensioned accordingly.

The magnetic design implemented using the ETD44/N87 core has been verified to operate within safe limits in terms of flux density variation and magnetizing current. On the output stage, the selected 68 μH inductor and 1000 μF capacitor values ensure low output voltage ripple and continuous conduction mode operation. The semiconductor stress and loss analyses demonstrate that the selected topology is suitable for this power level.

Simulation results indicate an output voltage of approximately 24.03 V and an output current of 15.02 A, thereby validating the design targets. Furthermore, the closed-loop control structure and the Type-III compensation network are observed to provide stable voltage regulation.

In conclusion, the designed two-switch forward converter is found to be successful in terms of both electrical performance and practical design feasibility. In future work, efficiency improvement through synchronous rectification, more detailed thermal analysis, and advanced modeling including parasitic effects may be carried out.

11. References

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